```
!git clone https://github.com/HelloShiwansh/Text-Summarizer
```

```
Cloning into 'Text-Summarizer'...
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 10 (delta 1), reused 10 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (10/10), 3.82 MiB | 4.12 MiB/s, done.
Resolving deltas: 100% (1/1), done.
```

```
%cd Text-Summarizer
```

```
/content/Text-Summarizer/Text-Summarizer
```

```
import os
print(os.listdir())
```

```
['samsum-validation.csv', 'text_summarizer.ipynb', '.gitignore', 'samsum-test.csv', 'samsum-train.csv', '.git']
```

```
import torch
print(torch.cuda.is_available())
```

```
True
```

```
import sys
print(sys.executable)
```

```
/usr/bin/python3
```

```
!pip install transformers
!pip install "transformers[torch]"
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transfo
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Requirement already satisfied: transformers[torch] in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers[t
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (26.0
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (20
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers[t
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (0
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (4.67.3)
Requirement already satisfied: torch>=2.2 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (2.10.0+cu
Requirement already satisfied: accelerate>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from transformers[torch]) (1.
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate>=1.1.0->transformers[tor
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transfo
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->transformers[torch]
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->transformers[tor
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->transformers[t
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->transformers[torch]) (3
```

```
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->transfor
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->t
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.2->t
```

```
import pandas as pd
from transformers import T5Tokenizer, T5ForConditionalGeneration, Trainer, TrainingArguments
```

```
train_data = pd.read_csv("samsum-train.csv")
val_data = pd.read_csv("samsum-validation.csv")
```

```
train_data.head()
```

	id	dialogue	summary
0	13818513	Amanda: I baked cookies. Do you want some?\nJ...	Amanda baked cookies and will bring Jerry some...
1	13728867	Olivia: Who are you voting for in this electio...	Olivia and Olivier are voting for liberals in ...
2	13681000	Tim: Hi, what's up?\nKim: Bad mood tbh, I was ...	Kim may try the pomodoro technique recommended...
3	13730747	Edward: Rachel, I think I'm in ove with Bella....	Edward thinks he is in love with Bella. Rachel...
4	13728094	Sam: hey overheard rick say something\nSam: i...	Sam is confused, because he overheard Rick com...

```
train_data.shape, val_data.shape
```

```
((14732, 3), (818, 3))
```

```
#random samping
train_data = train_data.sample(n=4000, random_state=42).reset_index(drop=True)
val_data = val_data.sample(n=500, random_state=42).reset_index(drop=True)
```

Data Pre-Processing

```
import re
```

```
def clean_data(text):
    text = re.sub(r'\s+', ' ', text)  # Replace multiple whitespace with a single space
    text = re.sub(r'\r\n', ' ', text)  # Replace newlines with a space
    text = re.sub(r"<.*?>", "", text)  # Remove HTML tags
    return text.strip().lower()
```

```
train_data['dialogue'] = train_data['dialogue'].apply(clean_data)
train_data['summary'] = train_data['summary'].apply(clean_data)

val_data['dialogue'] = val_data['dialogue'].apply(clean_data)
val_data['summary'] = val_data['summary'].apply(clean_data)
```

Tokenization

```
# using t5 tokenizzer to get the t5 vocab and tokenization
tokenizer = T5Tokenizer.from_pretrained('t5-small')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
tokenizer_config.json:  2.32k/? [00:00<00:00, 76.9kB/s]
spiece.model: 100%  792k/792k [00:00<00:00, 949kB/s]
tokenizer.json:  1.39M/? [00:00<00:00, 2.41MB/s]
```

```
#raw data -> tokenization -> model input
# for fine tuning
```

```
def tokenize(data):
    inputs = tokenizer(data['dialogue'], padding='max_length', truncation=True, max_length=512)
    targets = tokenizer(data['summary'], padding='max_length', truncation=True, max_length=150)

    inputs['labels'] = targets['input_ids']
    return inputs
```

```
train_dataset = train_data.apply(tokenize, axis=1).to_list()
val_dataset = val_data.apply(tokenize, axis=1).to_list()
```

```
train_dataset[0]
```

```
{'input_ids': [25208, 10, 7102, 55, 3, 23, 764, 640, 48, 403, 17, 77, 31, 7, 1108, 11, 3, 23, 816, 24, 25, 429, 253, 34,
1477, 25208, 10, 3, 7997, 15, 10, 7102, 55, 3, 10, 61, 2049, 6, 68, 3, 23, 31, 162, 641, 608, 34, 5, 3, 10, 61, 3, 7997,
15, 10, 68, 2049, 21, 1631, 81, 140, 3, 10, 61, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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'labels': [25208, 1622, 3, 7997, 15, 403, 17, 77, 31, 7, 1108, 5, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]}
```

```
# input_ids - token ids for the dialogue
# 1 is for EOS token and 0 is for padding

# attention_mask - mask to indicate which tokens are padding and which are not
# length of input_ids and attention_mask is same

# labels - token ids for the summary
```

```
len(train_dataset), len(val_dataset)
```

```
(4000, 500)
```

```
len(train_dataset[0]['input_ids']), len(train_dataset[0]['labels'])
```

```
(512, 150)
```

Working With the model

```
# NLP -> generation task which is dependent on the context of the input text -> so conditional generation task

model = T5ForConditionalGeneration.from_pretrained('t5-small')
```

```
config.json:          1.21k/?  [00:00<00:00,  33.3kB/s]
model.safetensors:  100%                                    242M/242M  [00:03<00:00,  194MB/s]
Loading weights:  100%                                      131/131  [00:00<00:00,  182.25it/s,  Materializing  param=shared.weight]
generation_config.json:  100%                               147/147  [00:00<00:00,  5.25kB/s]
```

```
import torch

if torch.backends.mps.is_available():
    device = torch.device("mps")
```

```
elif torch.cuda.is_available():
    device = torch.device("cuda")
else:
    device = torch.device("cpu")

print(f"Using device: {device}")
model.to(device)
```

```
Using device: cuda
T5ForConditionalGeneration(
  (shared): Embedding(32128, 512)
  (encoder): T5Stack(
    (embed_tokens): Embedding(32128, 512)
    (block): ModuleList(
      (0): T5Block(
        (layer): ModuleList(
          (0): T5LayerSelfAttention(
            (SelfAttention): T5Attention(
              (q): Linear(in_features=512, out_features=512, bias=False)
              (k): Linear(in_features=512, out_features=512, bias=False)
              (v): Linear(in_features=512, out_features=512, bias=False)
              (o): Linear(in_features=512, out_features=512, bias=False)
              (relative_attention_bias): Embedding(32, 8)
            )
            (layer_norm): T5LayerNorm()
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (1): T5LayerFF(
            (DenseReluDense): T5DenseActDense(
              (wi): Linear(in_features=512, out_features=2048, bias=False)
              (wo): Linear(in_features=2048, out_features=512, bias=False)
              (dropout): Dropout(p=0.1, inplace=False)
              (act): ReLU()
            )
            (layer_norm): T5LayerNorm()
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
      (1-5): 5 x T5Block(
        (layer): ModuleList(
          (0): T5LayerSelfAttention(
            (SelfAttention): T5Attention(
              (q): Linear(in_features=512, out_features=512, bias=False)
              (k): Linear(in_features=512, out_features=512, bias=False)
              (v): Linear(in_features=512, out_features=512, bias=False)
              (o): Linear(in_features=512, out_features=512, bias=False)
            )
            (layer_norm): T5LayerNorm()
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (1): T5LayerFF(
            (DenseReluDense): T5DenseActDense(
              (wi): Linear(in_features=512, out_features=2048, bias=False)
              (wo): Linear(in_features=2048, out_features=512, bias=False)
              (dropout): Dropout(p=0.1, inplace=False)
              (act): ReLU()
            )
            (layer_norm): T5LayerNorm()
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
    )
    (final_layer_norm): T5LayerNorm()
    (dropout): Dropout(p=0.1, inplace=False)
```

```
# Training Arguments

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=6,
    weight_decay=0.01,

    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,

    eval_strategy="epoch",
    save_strategy="epoch",

    warmup_steps=500
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
```

```
    train_dataset=train_dataset,
    eval_dataset=val_dataset
)
```

```
# trainning the model
trainer.train()
```

[3000/3000 24:07, Epoch 6/6]

Epoch	Training Loss	Validation Loss
1	3.577381	0.380097
2	0.397485	0.359731
3	0.374499	0.354865
4	0.362143	0.350373
5	0.355740	0.349276
6	0.350639	0.349105

```
Writing model shards: 100%                                    1/1 [00:00<00:00,  1.43it/s]
Writing model shards: 100%                                    1/1 [00:07<00:00,  7.70s/it]
Writing model shards: 100%                                    1/1 [00:12<00:00, 12.38s/it]
Writing model shards: 100%                                    1/1 [00:34<00:00, 34.40s/it]
Writing model shards: 100%                                    1/1 [00:36<00:00, 36.64s/it]
Writing model shards: 100%                                    1/1 [00:32<00:00, 32.22s/it]
TrainOutput(global_step=3000, training_loss=0.9029809265136719, metrics={'train_runtime': 1450.2115,
'train_samples_per_second': 16.549, 'train_steps_per_second': 2.069, 'total_flos': 3248203235328000.0, 'train_loss':
```

```
# model load => fine tunning => Trainning => Saving the model
```

```
model.save_pretrained("./saved_summary_model")
tokenizer.save_pretrained("./saved_summary_model")
```

```
Writing model shards: 100%                                    1/1 [00:47<00:00, 47.75s/it]
('./saved_summary_model/tokenizer_config.json',
 './saved_summary_model/tokenizer.json')
```

```
!pip install huggingface_hub

from huggingface_hub import notebook_login
notebook_login()
```

```
Requirement already satisfied: huggingface_hub in /usr/local/lib/python3.12/dist-packages (1.10.1)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (3.25.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (1.4.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (0.28.1)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (6.0.3)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (4.67.3)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (4
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub) (4.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub) (
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub) (3.1
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub) (8.3.2)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub) (
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub) (13.9.4
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
```

```
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer

model = AutoModelForSeq2SeqLM.from_pretrained("saved_summary_model")
tokenizer = AutoTokenizer.from_pretrained("saved_summary_model")

model.push_to_hub("HelloShiwansh/text-summarizer-model")
tokenizer.push_to_hub("HelloShiwansh/text-summarizer-model")
```

```
Loading weights: 100%|| 131/131 [00:00<00:00, 425.71it/s, Materializing param=shared.weight]
Writing model shards: 100%|| 1/1 [00:06<00:00, 6.33s/it]
Processing Files (1 / 1)      : 100%|| 242MB / 242MB, 67.2MB/s
New Data Upload               : |          0.00B / 0.00B, 0.00B/s
  ...0zi1gka/model.safetensors: 100%|| 242MB / 242MB
README.md: 5.17k/? [00:00<00:00, 310kB/s]
CommitInfo(commit_url='https://huggingface.co/HelloShiwansh/text-summarizer-model/commit/f14b7e49e0df8227d73cc3020c91f8d70700182f', commit_message='Upload tokenizer', commit_description='', oid='f14b7e49e0df8227d73cc3020c91f8d70700182f', pr_url=None, repo_url=RepoUrl('https://huggingface.co/HelloShiwansh/text-
```

```
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer

model = AutoModelForSeq2SeqLM.from_pretrained("HelloShiwansh/text-summarizer-model")
tokenizer = AutoTokenizer.from_pretrained("HelloShiwansh/text-summarizer-model")
```

```
Loading weights: 100%|| 131/131 [00:00<00:00, 269.84it/s, Materializing param=shared.weight]
```

Testing

```
def summarize_dialogue(dialogue):
  dialogue = clean_data(dialogue)

  #tokenise
  inputs = tokenizer(
      dialogue,
      max_length=512,
      padding='max_length',
      truncation=True,
      return_tensors='pt'
  ).to(device)

  #generate the summary (token ids)
  model.to(device)
  target = model.generate(
      input_ids = inputs['input_ids'],
      attention_mask = inputs['attention_mask'],
      max_length=150,
      num_beams=4,
      early_stopping=True
  )

  #decoding(ids -> text)
  summary = tokenizer.decode(target[0], skip_special_tokens=True)

  return summary
```

```
test_dialogue = """
Journalist: Climate change is becoming one of the most pressing issues worldwide, affecting ecosystems and human lives.

Scientist: Yes, rising global temperatures are causing melting glaciers, rising sea levels, and more extreme weather events

Journalist: Governments are trying to implement policies, but progress seems slow.

Scientist: That’s true. Transitioning to renewable energy and reducing carbon emissions requires global cooperation and lor

Journalist: What role can individuals play in this situation?

Scientist: Individuals can contribute by reducing energy consumption, using public transport, and supporting sustainable pr

Journalist: Are there any positive developments?

Scientist: Absolutely. Advances in solar and wind energy, along with increased awareness, are promising signs.

Journalist: So, there is still hope?

Scientist: Yes, but immediate and consistent action is crucial to mitigate long-term damage.
"""
```

```
summary = summarize_dialogue(test_dialogue)
print(summary)
```

```
climate change is becoming one of the most pressing issues worldwide, affecting ecosystems and human lives.
```

```
Start coding or generate with AI.
```